



A Framework for the Validation and Visualization of Numeric Control (NC) Data for Autonomous Robotic Drilling

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Manufacturing Engineering Dept

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TO WHOMSOEVER IT MAY CONCERN

Training completion Letter

This is to certify that **Ms.Samriddhi Mishra** has successfully completed her training in **Manufacturing Engineering** department at TATA Boeing Aerospace Ltd. Hyderabad from **12-May-2025 to 09-Aug-2025**.

We appreciate her valuable contribution and wish her all the very best in her future endeavours.

For TATA Boeing Aerospace limited


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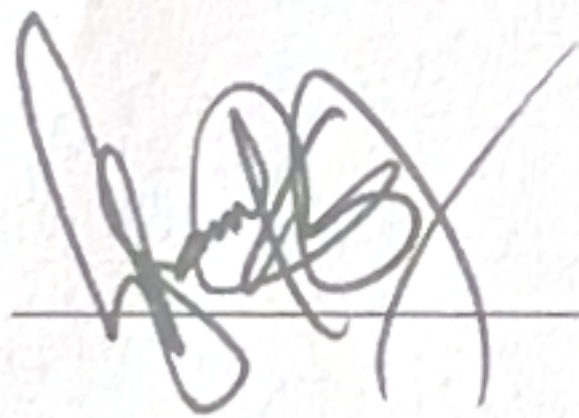
I, Samriddhi Mishra, hereby declare that this internship report titled "A Framework for the Validation and Visualization of Numeric Control (NC) Data for Autonomous Robotic Drilling" is a record of the original work undertaken by me as part of my internship pursued at Tata Boeing Aerospace Limited, Hyderabad from May 12, 2025, to August 9, 2025.

The work submitted is a result of my own effort and has not been submitted to any other institution for the award of any degree or diploma. All the information provided in this report is authentic to the best of my knowledge.



Samriddhi Mishra

This is to certify that the above statement made by the student is correct to the best of our knowledge.



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(Internship Supervisor)

Abstract

In an aerospace manufacturing environment, precision is paramount. Hence, the verification of Numeric Control (NC) data for autonomous robots is critical to prevent costly errors. This project addresses the "Black Box Problem" associated with the robotic drilling process of the 737 Vertical Fin at Tata Boeing Aerospace Limited (TBAL), where the sheer amount of data makes manual verification inefficient. The primary objective was to develop a streamlined, web-based framework to validate and visualize this data. The solution is delivered through a central web portal that guides users to two distinct, browser-based applications. The Comparison tool allows engineers to upload consolidated manufacturing data and generates an interactive 3D plot highlighting discrepancies between engineering specifications and actual NC data. The Visualization tool provides a detailed 3D model of the robot's drill paths for user-selected operations, also generated directly within the browser. By centralizing access and leveraging in-browser computation with Python via WebAssembly (Pyodide), the framework significantly reduces verification time, increases accuracy, and provides an easily accessible data validation resource for the engineering team.

Acknowledgements

I express my deepest gratitude to the amazing 737 Manufacturing Engineering team at Tata Boeing Aerospace Limited (TBAL) for always making me feel like a part of the team and being patient with my countless questions and mistakes. The opportunity to work alongside such a dedicated and welcoming team was an experience that I will cherish forever.

I want to extend special thanks to **Mr. Chirag Goyal Sir** and **Mr. Dayaan Zargar Sir**. Thank you for trusting me with this project and for believing in my abilities, especially in the moments where I doubted myself. Your mentorship and guidance were instrumental for the project and contributed significantly to my personal growth as an engineer.

Finally, I would like to thank **Mr. Naveen Sunkoju Sir** for his exceptional leadership and the entire staff at TBAL for fostering an innovative and collaborative environment. The opportunity to work at such a state-of-the-art manufacturing facility was truly an unparalleled experience.

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1. Introduction

1.1 Overview of Tata Boeing Aerospace Limited (TBAL)

Tata Boeing Aerospace Limited (TBAL) is a Joint Venture (JV) between The Boeing Company (USA) and Tata Advanced Systems Limited (India). It is a state of the art assembly facility that is responsible for the assembly of the Boeing 737 MAX Vertical Fins and it is the global single source of fuselages for the AH-64 Apache Helicopters (Tata-Boeing JV Establishes Aerospace Facility in Hyderabad, 2018).

1.2 The Boeing 737 Vertical Fin Assembly

The Boeing 737 is one of the most commercially successful aircraft lines in history and is known for its versatility and efficiency. The TBAL JV deals with the assembly of the vertical fin for Boeing 737 MAX planes. The detailed parts are sourced from manufacturing facilities like Tata Advanced Systems Limited (TASL) and are then assembled in TBAL using a pulse line manufacturing approach. In this approach, the entire assembly line moves forward simultaneously from one position to another in discrete steps or “pulses”. This method helps establish a predictable production rhythm and helps standardize the work cycles. It also helps in easier identification of bottlenecks at different stages compared to a continuously moving assembly line.

1.3 Manufacturing Engineering

The Manufacturing Engineering department at TBAL forms a crucial link between design and production. It is responsible for planning and execution of the entire manufacturing process of the AH-64 fuselage and the B737 MAX vertical fin. The skillset needed for this task consists of strong engineering principles and production methodologies. In the context of the Vertical Fin, manufacturing engineers at TBAL are responsible to ensure that the pulse line manufacturing approach runs smoothly and efficiently.

A key function of this department is to create Manufacturing Instruction Sheets (MISs) that provide the production department with clear step by step manufacturing instructions. In the context of automated manufacturing, this role translates to converting the instructions into a language that the machines can comprehend (i.e. coding).

1.4 The Role of Automation

Automation plays a crucial role in manufacturing facilities. It helps achieve the core objectives of manufacturing: producing higher quality products in a short time at a lower cost and with increased safety. It also ensures high precision which is crucial in the Aerospace sector because of tight tolerances. I had the privilege of learning about manufacturing automation at TBAL by observing the KUKA KR_300_R2700-2_V00 autonomous industrial drilling robots in action. Drilling and Countersink are critical operations in the Vertical fin assembly and require high precision. The robot accomplishes this efficiently with minimal manual supervision using its 6 axis coordinate system. The first step is the drilling which happens separately for the Left Hand (LH) and Right Hand (RH) sides of the Vertical Fin. There are 3 drilling cycles for each side. The drilling operation is followed by the countersink operation for both sides. The holes that are deemed to be unsuitable for robotic drilling, perhaps because of a complex surface curve, are left out during these operations and drilled manually afterwards.

1.5 Project Objective

While the robot significantly enhances efficiency and precision, it has complex Numeric Control (NC) codes that work in the backend to guide the robot. These codes contain ~100,000 lines and are difficult to verify manually.

Therefore, the primary objective of this project was to design a software framework for comprehensive validation and visualization of the robotic drilling processes. My main goals were

- to create an automated validation tool that automatically parses NC and Program files, consolidating key drilling parameters (coordinates, drill length, cycle time etc.) onto a single sheet for direct side-by-side comparison,
- to create an interactive 3D visualization model that translates the abstract coordinate data from the NC files into an interactive 3D representation of the Vertical Fin, and
- to ensure geometric accuracy by converting the robot's local coordinate system to the part's absolute coordinate system for accurate real-world representation.

Ultimately, the goal was to create a framework that helps validate the NC data and improve the overall efficiency of the manufacturing engineering team at TBAL.

2. The Challenge: Ensuring Fidelity in Robotic Manufacturing

2.1 NC Programming

As mentioned in section 1.4, the autonomous drilling robot needs to be fed NC Codes that program how the robot moves. The particular robot at TBAL utilises SINUMERIK 840D SL documentation for NC Programming which is one of the most advanced programming systems used for Computer Numeric Control (CNC) machines. A typical NC Program consists of coordinates for the absolute, top, and bottom reference points for the drill points as well as the approach point coordinates and the pitch, roll, and yaw rotation of the robot's End Of Arm Tool (EOAT). Approach and reference points are not drilled, rather they are used to describe the path the robot takes to get to a drill point. These NC programs are made for each drill operation that the robot performs. A master program is then compiled where all drilling operations are called in a specified sequence.

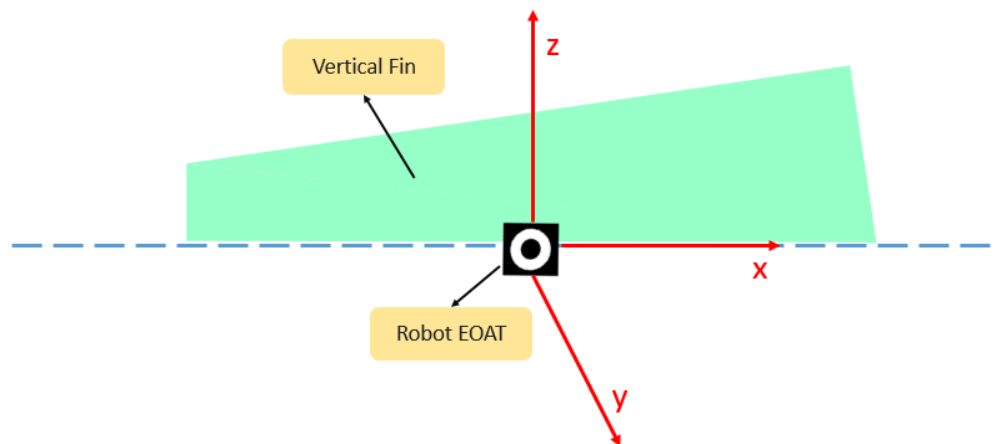


Figure 1: A simplified diagram illustrating the local coordinate system of the Robot's End of Arm Tool (EOAT) relative to the global coordinate system of the Vertical Fin

2.2 The “Black Box” Problem

NC programs contain all the crucial data for the robot to function. However, the lack of visualization and the sheer

amount of data present makes the codes quite difficult to comprehend. Because NC Programming is not transparent, we are presented with a “Black Box Problem”. We can observe the inputs and the outputs, but the process behind it is hidden. This lack of visualization makes it difficult to spot errors in path and orientation of the robot.

2.3 The Need for Pre-Process Verification

Given the nature of NC programs as explained in section 2.2 and the high stakes and tight tolerances in an aerospace environment, the cost of producing an error is significantly high and can potentially lead to scrapping of the entire part. Hence, it is not advisable to assume that any NC code is free of errors. This implies that pre-process verification is a crucial step that cannot be taken lightly. We need tools that engineers can use to validate the data and visualize the robot’s path before the NC codes are fed. Moreover, these tools need to be straightforward enough so that any engineer without coding knowledge can access them using easily available resources.

3. Development of the Validation and Visualization Framework

3.1 Framework Architecture

To address the challenges of the “Black Box Problem”, a web-based framework was designed. The objective was to provide separate tools for data validation and visualization that are interconnected by a single entry point.

The single entry point is a Google site that has buttons to direct the user to the tools for visualization and validation. The site presents two buttons: Compare and Visualize. Each option directs the user to a page which is HTML-embedded to perform the respective operation.

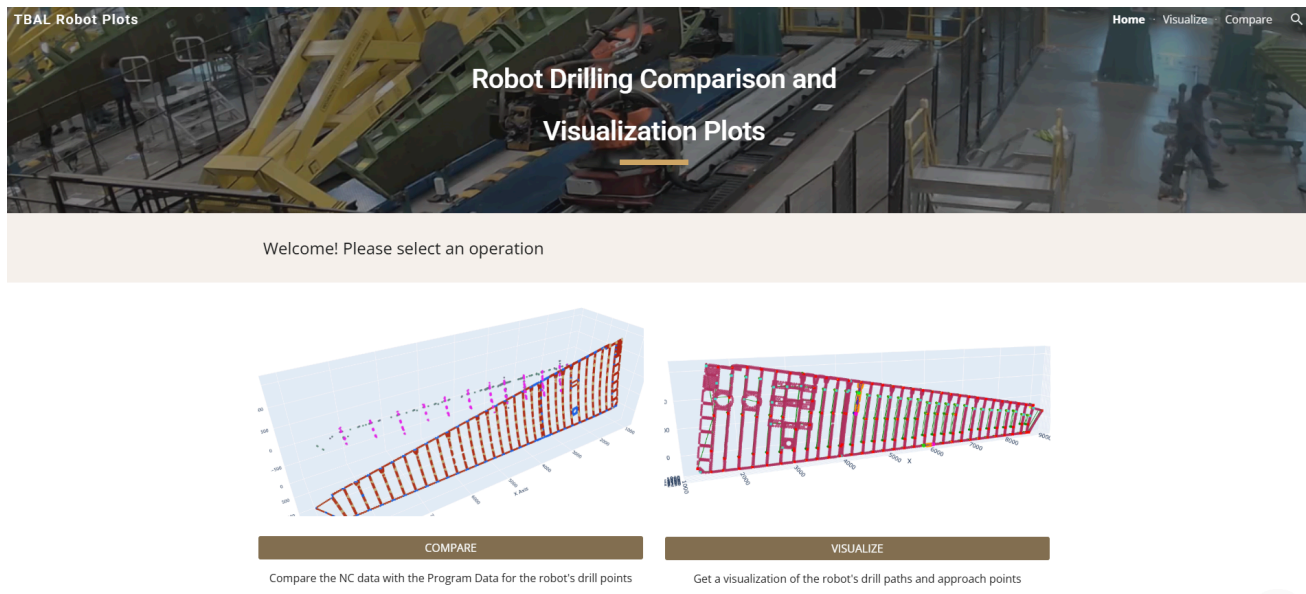


Figure 2: The main web portal, built with Google Sites, serving as a central access point for the Comparison and Visualization applications.

Firstly, a Data Consolidation Model was built in Microsoft Excel using Visual Basic for Applications (VBA). Its function is to parse the raw, complex NC Data files and master Program Files, extracting and structuring the values

of critical parameters onto a single sheet for analysis. This file acts as a prerequisite for the website.

	Column1	CY1	CY2	CY3
RUN	Program	C:\Users\cgoyal\Documents\RH_CY1.WPD	C:\Users\cgoyal\Documents\RH_CY2.WPD	C:\Users\cgoyal\Documents\RH.WPD
	GD6	C:\Users\cgoyal\Documents\NC_GD6 1.INI	C:\Users\cgoyal\Documents\NC_GD6 2.INI	C:\Users\cgoyal\Documents\NC_GD6 3.INI
GENERATE PDF	merge file	C:\Users\cgoyal\Documents\merge_files.bat		

Figure 3: The user interface of the Excel VBA tool designed to parse and consolidate raw NC and Program data into a single, structured file for analysis.

The core of the framework consists of two powerful, self-contained HTML applications that run Python logic directly in the browser using WebAssembly (via the Pyodide project):

- The Comparison Application takes the consolidated Excel sheet as an input to perform a validation check.
- The Visualization Application uses the same consolidated sheet to generate a 3D model of the drilling process.

This architecture ensures that the tools are easily accessible to anyone who has access to a browser by eliminating the need to download specific coding libraries.

3.2 The Validation Module: An automated Data Parser & Comparator

The primary objective of the Validation Module is to automate the parsing of complex NC data and compare their parameters with the master engineering specifications. It works in the following steps:

1. Access and Upload: The engineer navigates to the 'Comparison' page on the central Google Site. The embedded HTML application prompts the user to upload the consolidated Excel sheet generated by the VBA data parser.

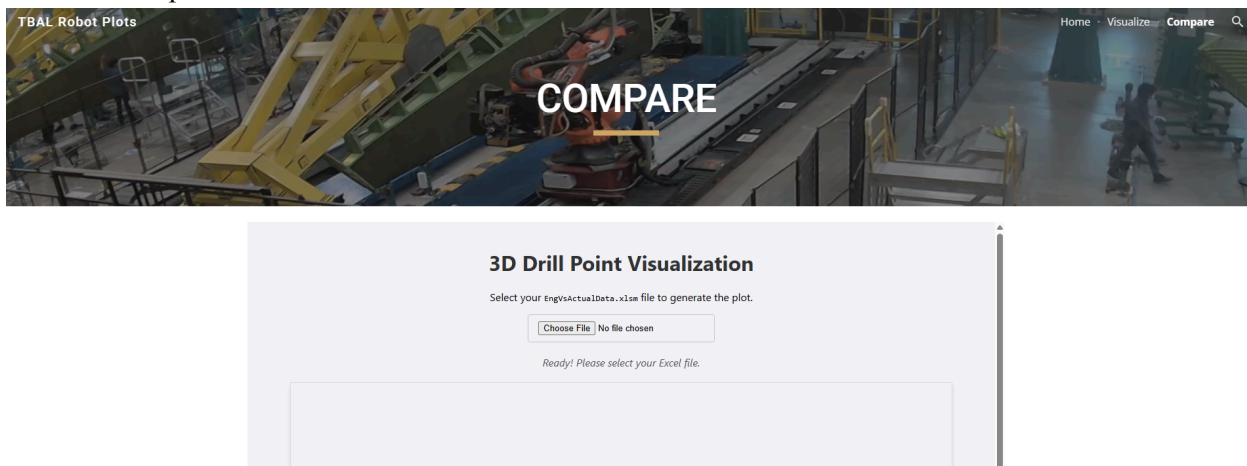


Figure 4: The 'Comparison' application interface, prompting the user to upload the consolidated data sheet to generate a 3D discrepancy plot.

2. **Outlier Identification:** Once the file is uploaded, the application uses Pyodide (a WebAssembly port of Python) to execute Python, Pandas, and Plotly logic directly and securely within the browser. This generates an interactive 3D plot that visually highlights discrepancies between the engineering specifications and the actual NC data.

3.3 The Visualization Module: An Interactive 3D Model

The primary objective of the Visualization Module is to translate the abstract coordinate system of the NC programs into an interactive 3D model. This offers a solution to the “Black Box Problem” by allowing engineers to visualize the drill paths and robot orientation. This is achieved via the 'Visualization' application on the project portal.

The module operates in the following steps:

1. **User Input:** After navigating to the 'Visualization' page, the user is prompted to upload the consolidated data sheet. The interface also provides input fields for the user to specify the list of drilling operations they wish to visualize.
2. **3D Plot generation:** The application's embedded Python script parses the uploaded data to extract the X, Y, and Z coordinates for the drill points and approach points corresponding to the user-selected operations. It then uses the Plotly library to generate a comprehensive 3D plot displaying the drill points and paths.

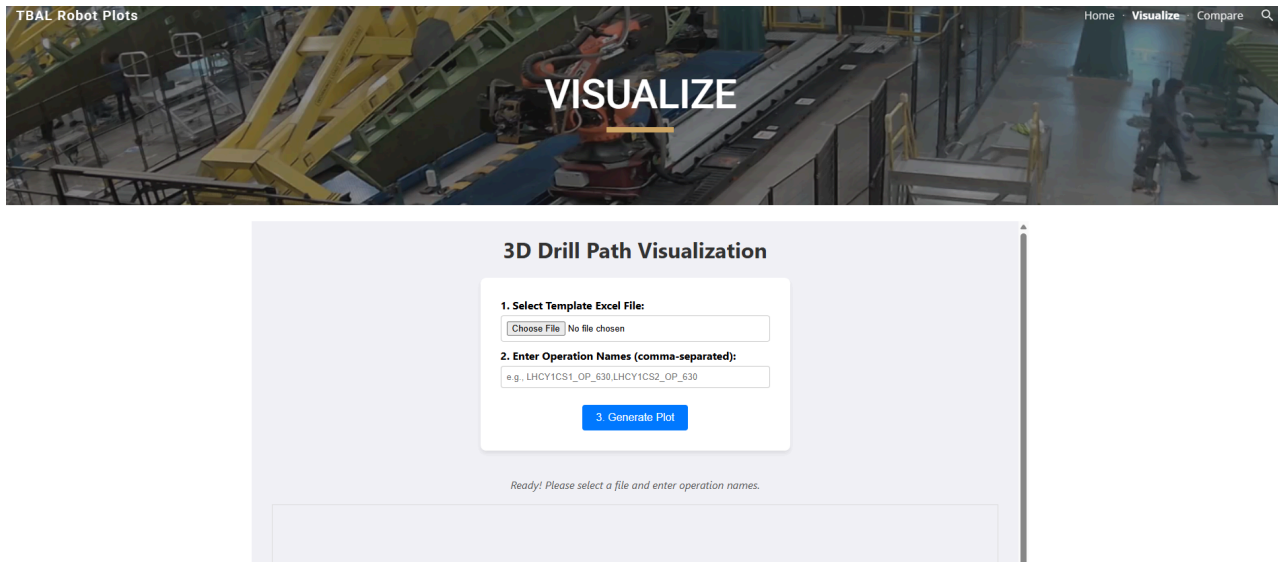


Figure 5: The 'Visualization' application interface, where users upload the data sheet and specify operation names to render a 3D model of the drill paths.

3.4 Ensuring Geometric Accuracy: The Coordinate System Transformation

Although the visualization tool was useful in creating the structure of the vertical fin and locating the drill points, a key fallacy was observed: the rotation of the EOAT around the y axis was not being taken into account. This materialised as a mismatch between the global coordinate system of the stationary vertical fin and the local coordinate system of the robot's EOAT.

To correct the misalignment, a coordinate transformation was implemented using the standard y - rotation matrix to convert the local coordinates to global coordinates. The equations derived from the rotation matrix transformation were

$$x' = x \cos(b) - z \sin(b)$$

$$y' = y$$

$$z' = (z + x' \sin(b)) / \cos(b)$$

Where x' , y' , and z' are the absolute coordinates and x , y , and z are the coordinates in the rotated frame of reference.

This approach resulted in a more accurate visualization of the Vertical Fin and made the model more reliable. However, the approach still has some discrepancies that need more research.

3D Plot of All Drill Points with Original and Rotated References

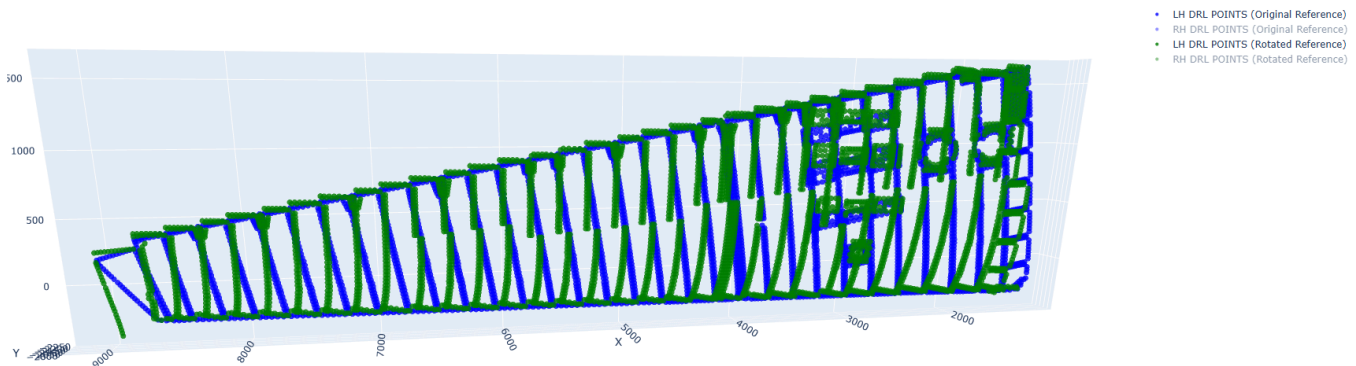


Figure 6: A 3D plot demonstrating the effect of the coordinate transformation, showing the original reference points (blue) and the transformed points (green) in their proper geometric orientation.

4. Framework in Action: Results and Impact

4.1 Output

The developed framework delivers two sets of outputs from its main modules, providing both quantitative validation and qualitative visualization.

The Comparison Application takes a consolidated Excel sheet and produces an interactive 3D plot that visually flags discrepancies between the Program data and the actual NC data. Outlier points are clearly visible, providing an immediate quantitative validation output.

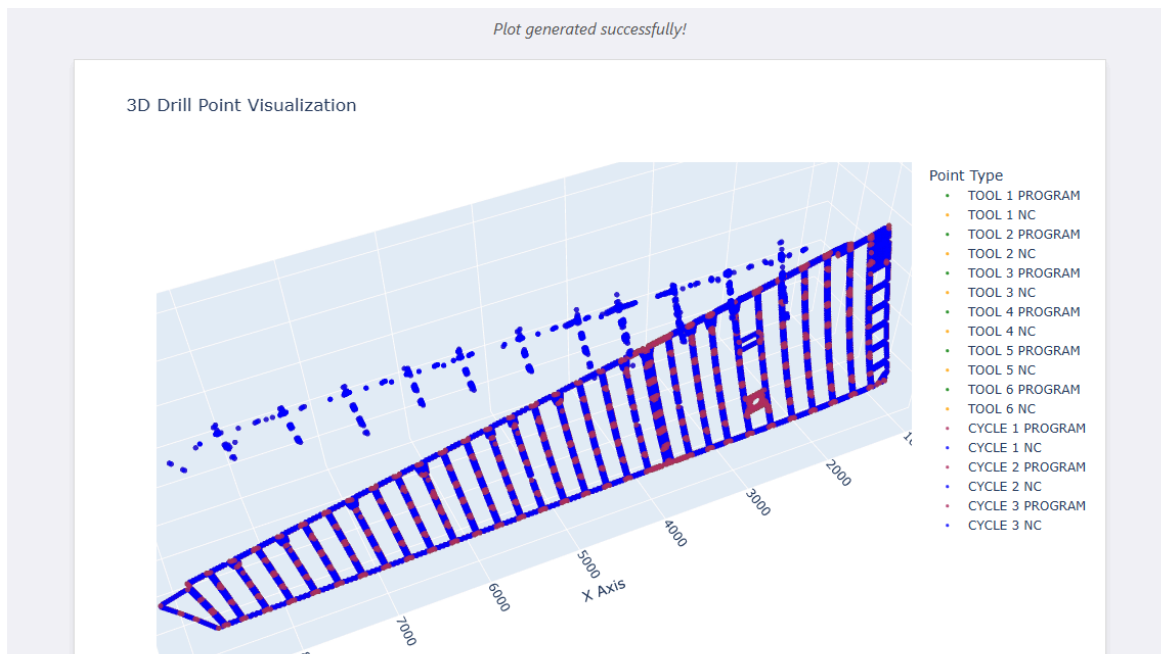


Figure 7: Output of the Comparison tool, where red points are indicative of the program data and blue points represent the NC data. The NC points that do not fall on the VFin are the outliers.

The Visualization Application produces a detailed, interactive 3D model of the Vertical Fin's drilling operations. Based on user-selected operations, this model plots the precise drill points, drill paths, and reference points, offering a clear qualitative view of the robot's intended paths.

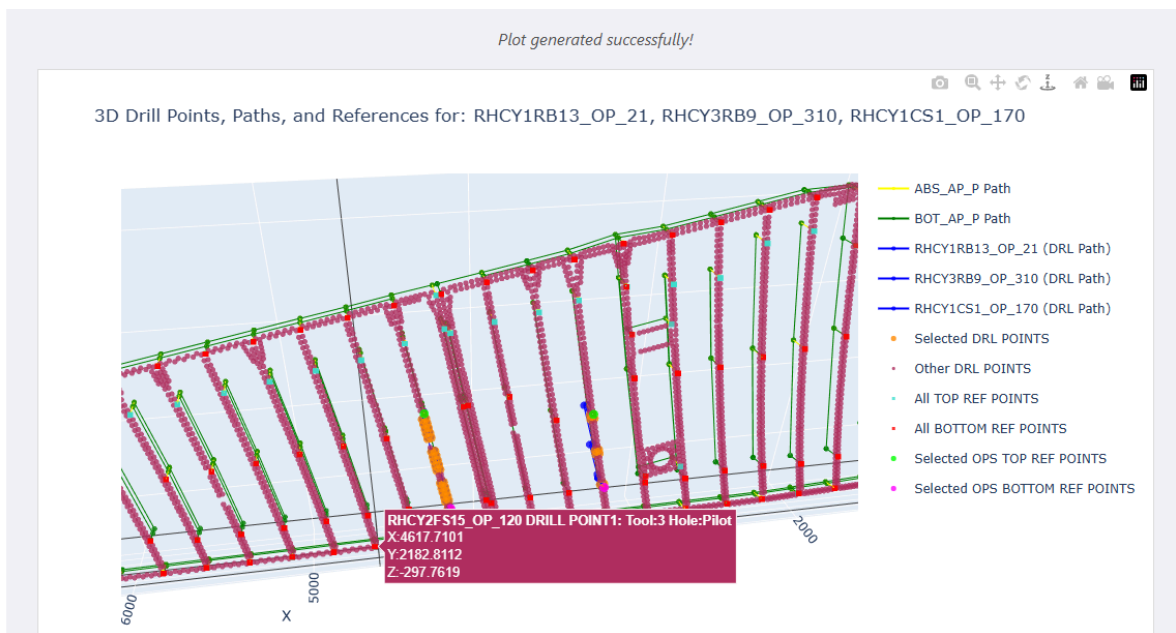


Figure 8: A detailed output from the Visualization tool, showing the rendered drill paths for selected operations. The hover-box provides specific coordinate and tool data for individual points.

4.2 Analysis of Discrepancies

As shown in Figure 7, the comparison model revealed discrepancies between the engineering specifications (Program Data) and the final robotic execution (NC Data). An analysis of these outliers revealed an inconsistent mapping procedure across different drilling cycles.

The comparison model was comparing two sets of data:

- The Program Data dataset dictates the intended location for all drill points. The specified program x,y, and z coordinates must be added to the respective reference points to get the true position of a drill point.
- The NC Data dataset represents the actual location where the holes were drilled. These coordinates are absolute and do not require the addition of reference points.

The comparison model was built on these assumptions. However, a deeper look into the outliers revealed a significant flaw in certain drilling cycles. For these specific cycles, a correct mapping between the intended and actual drill points could only be achieved by adding the reference point coordinates to both the Program data and the NC data. Furthermore, it was discovered that one of the drilling cycles on the Right Hand (RH) side contained programs that were missing a critical value needed to link the NC data back to the program data.

This inconsistency in the data processing logic is the primary reason the comparison tool flagged certain points as outliers, providing a crucial insight into a previously unnoticed process variation. This discovery, made possible through the framework's automated comparison, provides a clear path for process correction and prevents potential non-conformances that would have been difficult to trace manually.

4.3 Impact

The implementation of this Framework has a positive impact on the Manufacturing Engineering Department of TBAL. It enables proactive process correction for the robotic drilling operations. The framework helps flag errors and identify specific unnoticed anomalies in the current NC data logic in a minimal amount of time.

The key benefits of the framework include:

- Increased Efficiency: The team no longer needs to verify multiple ~100,000 line code files manually to flag errors. The automated process takes under 10 minutes to complete and frees up the team's valuable time for other engineering tasks.
- Enhanced Process Understanding: The visual representation of the robot's drilling operations enables the engineering team to get a better understanding of how the process is undertaken.
- Increased engineering confidence: With a reliable and accessible tool to validate and visualize the robot's actions, engineers can have much greater confidence in the autonomous drilling process. It also helps to clarify any misconceptions about the current process.

5. Conclusion and Learnings

5.1 Key Learnings

The opportunity to intern at a prestigious manufacturing facility like Tata Boeing Aerospace Limited (TBAL) was an immersive experience that taught me a lot about technical skills and professional engineering practices. The

project presented a steep learning curve, especially for someone like me with minimal prior coding experience. A critical challenge that offered the biggest growth opportunity was the initial runtime of the Validation module, which took over an hour to process the large NC data files. Researching optimization techniques for the code was a crucial learning experience for me and gave me a good understanding of computational performance. Eventually, I was able to get the runtime under 5 minutes.

Another significant challenge was integrating the 3D plot with the excel data for the validation module. I initially created a separate excel macro sheet to extract the data and a python code to plot it. I realised the plotting process needed to be more user-friendly. This motivated me to research how to store the extracted data locally in the browser and generate the plot there itself. The experience made me realise that it is important for a creation to be easily accessible and straightforward in its design.

Beyond the technical aspects, my internship helped me gain valuable practical insight into a state-of-the-art aerospace manufacturing environment. I gained a firsthand understanding of pulse line manufacturing methodologies, the critical role of automation in achieving high-precision drilling, and the complexities of NC programming for industrial robots. Ultimately, the project was a comprehensive lesson in practical problem-solving, from designing a modular framework to addressing the real-world "Black Box Problem" of NC data verification.

5.2 Future Scope

The Framework could benefit immensely by future research. The two main areas of focus could be System Integration and Geometric Accuracy.

Currently, the framework uses a central portal to direct users to two separate applications. A future iteration could merge these into a single-page application. In this integrated environment, a user could upload their data once and then toggle between the 'Comparison' and 'Visualization' views, creating a workflow without interruptions. Moreover, as mentioned in section 3.4, the coordinate transformation logic can be improved to obtain the accurate real world representation of the robot's drill points.

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